Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: GroundTruth
b. Team leader name: Lokesh Kadyan
c. Problem Statement: Sustainability & Social Impact: crop-loss insurance claims for smallholder farmers settled in minutes, not months, with satellite-verified evidence

Brief about the idea

Every season, floods, droughts and typhoons destroy smallholder crops across Asia-Pacific. Crop insurance exists, but claims wait weeks to months for manual field surveys, disputes are common, and fraud raises premiums for everyone.

GroundTruth lets a farmer file a claim with what they already have: a phone photo and a short note in their own language. Four agents on Google Cloud check it against satellite vegetation and rainfall evidence and settle it under the scheme rules in under a minute.

Principle: the AI explains; a deterministic engine proves. Gemini reads, translates and explains; every number and verdict comes from auditable rules, with clauses cited.

Opportunities

- How different is it from any of the other existing ideas?
- How will it be able to solve the problem?
- USP of the proposed solution

Different: Existing tools either digitise the claim form or run a black-box damage model. GroundTruth combines the farmer's own evidence with remote-sensing proof, and keeps the decision deterministic and cited.

Solves it by: Removing the survey from the critical path for corroborated claims. Surveys are reserved for the claims the data contradicts, so assessors spend their time where it matters.

USP: A dramatic photo cannot outrun the satellite: assessed loss is capped by what remote sensing supports. Fraud is flagged, not silently paid or silently rejected. Works in the farmer's language.

List of features offered by the solution

Multilingual multimodal intake Gemini reads the field photo + note in Hindi, Marathi, Filipino, English and more

Satellite + weather corroboration Vegetation-index drop, rainfall anomaly and standing-water days per district

Deterministic adjudicator Covered peril, 7-day intimation, 33% threshold, corroboration, payout formula

Loss cap Assessed loss never exceeds what the satellite supports

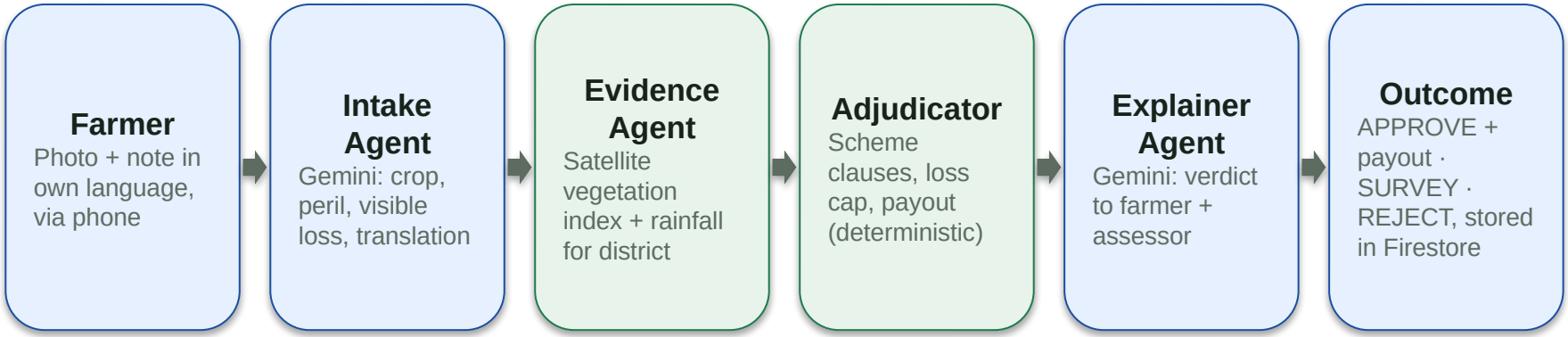
Fraud / mismatch flag Contradicted claims routed to physical survey

Explainable verdicts Farmer message in their language + assessor summary with cited clauses

Claims ledger Every claim, evidence pack and agent trace stored in Firestore

Never breaks Deterministic fallback if the model is unavailable

Process flow diagram or Use-case diagram

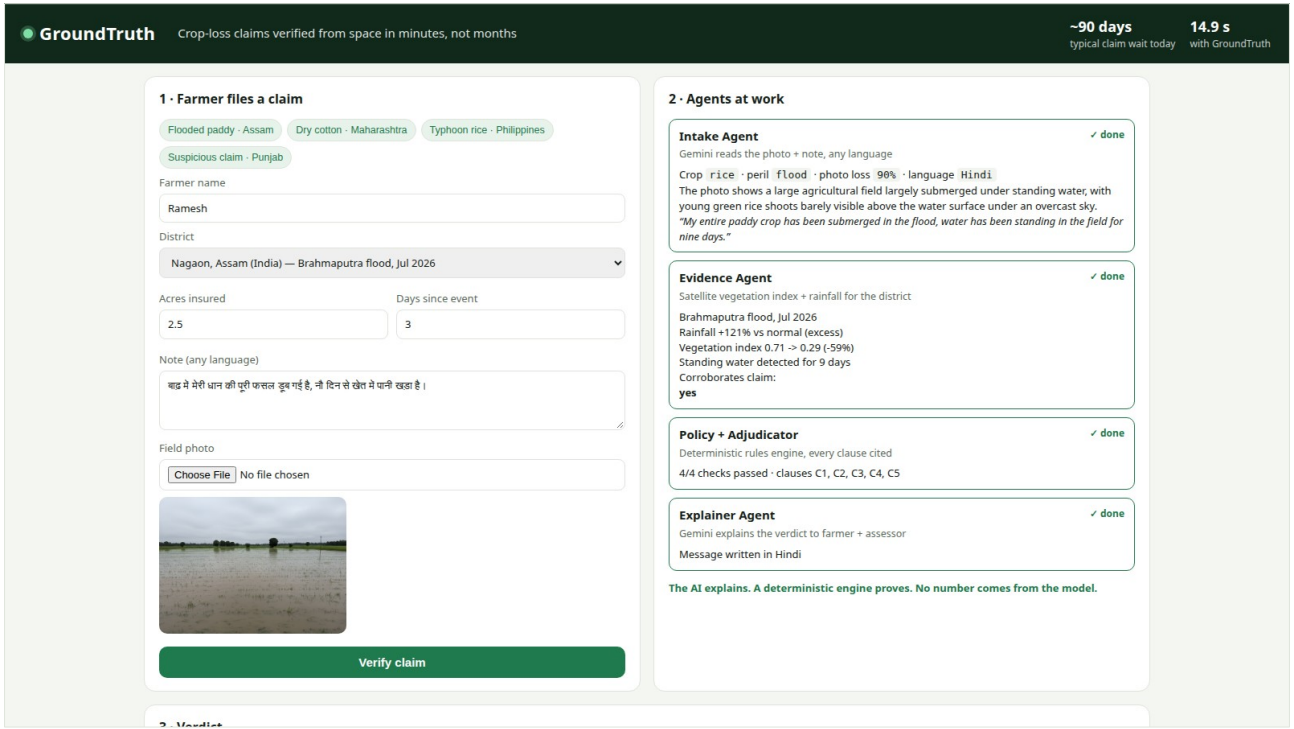


Blue = Gemini (reads, translates, explains)

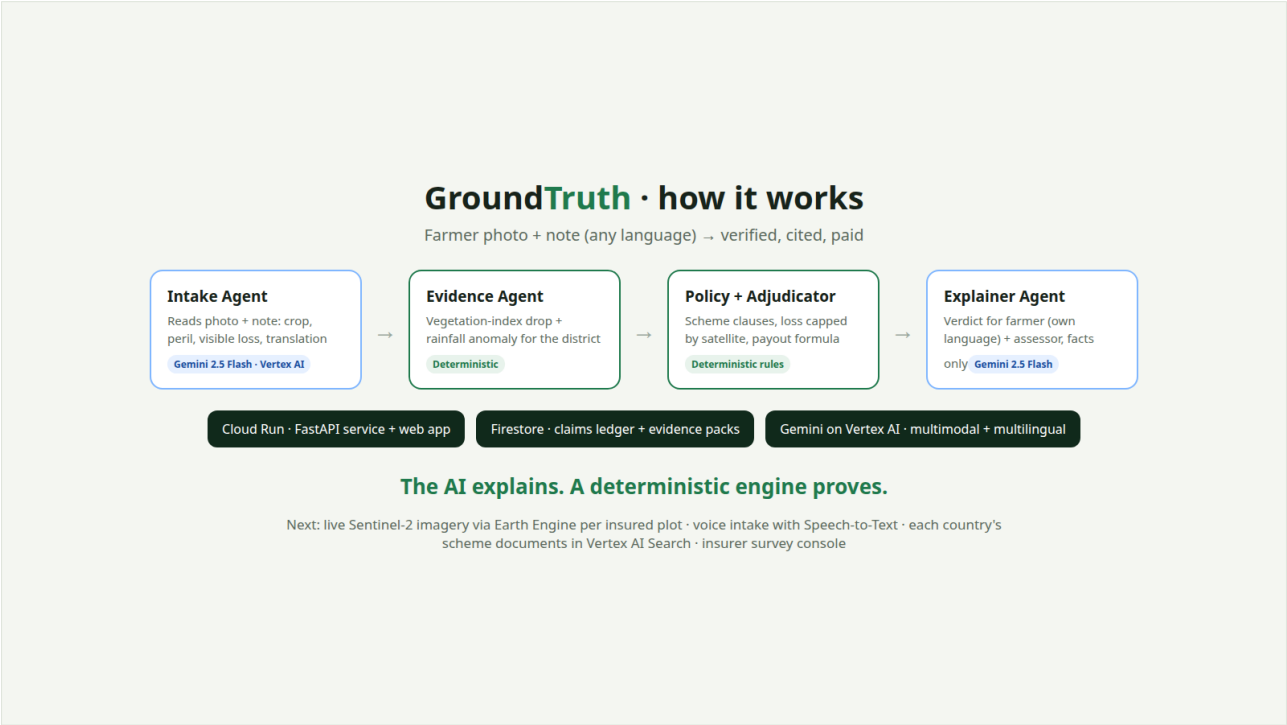
Green = deterministic engine (every number, every verdict)

Branch: photo/note contradicted by satellite + rainfall → fraud flag → physical survey queue

Wireframes/Mock diagrams of the proposed solution (optional)



Architecture diagram of the proposed solution



Technologies to be used in the solution

Gemini 2.5 Flash on Vertex AI multimodal intake agent and multilingual explainer agent

Cloud Run serverless FastAPI service + web app, deployed from source

Firestore (Firebase) claims ledger: claim, evidence pack, decision, agent trace

Cloud Build + Artifact Registry container build and image storage

Cloud Text-to-Speech narration for the demo

Python, FastAPI, vanilla JS no heavy front-end framework; accessible, mobile-friendly UI

Roadmap on GCP Earth Engine (Sentinel-2 per plot), Speech-to-Text voice intake, Vertex AI Search over scheme documents, BigQuery for portfolio analytics

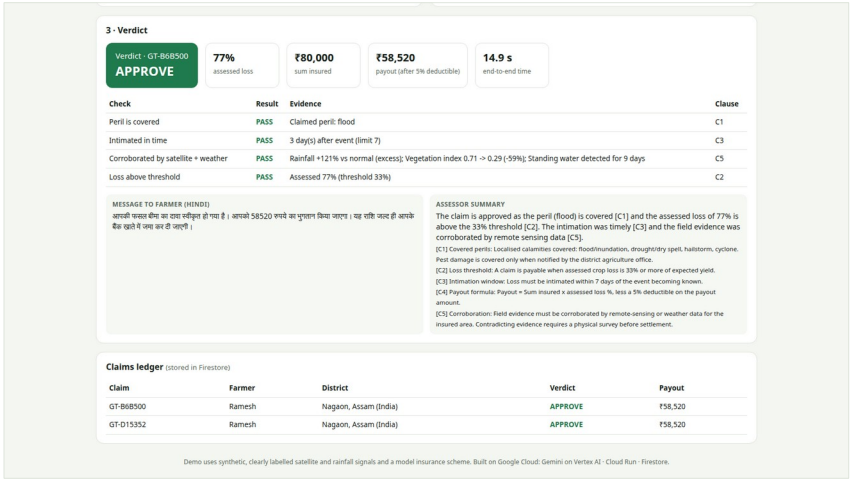
Estimated implementation cost (optional)

Per claim: 2 Gemini Flash calls (one with an image) + one Firestore write: well under one US cent per claim at list prices.

Platform: Cloud Run scales to zero; a pilot district of 10,000 claims a season runs for tens of US dollars of cloud cost.

Biggest saving: field surveys avoided for corroborated claims, which today cost insurers far more per claim than the cloud does.

Snapshots of the prototype



3 · Verdict

Verdict · GT-B6B500
APPROVE

77% assessed loss
₹80,000 sum insured
₹58,520 payout (after 5% deductible)
14.9 s end-to-end time

Check	Result	Evidence	Clause
Peril is covered	PASS	Claimed peril: flood	C1
Intimated in time	PASS	3 day(s) after event (limit 7)	C3
Corroborated by satellite + weather	PASS	Rainfall +121% vs normal (excess); Vegetation index 0.71 -> 0.29 (-59%); Standing water detected for 9 days	C5
Loss above threshold	PASS	Assessed 77% (threshold 33%)	C2

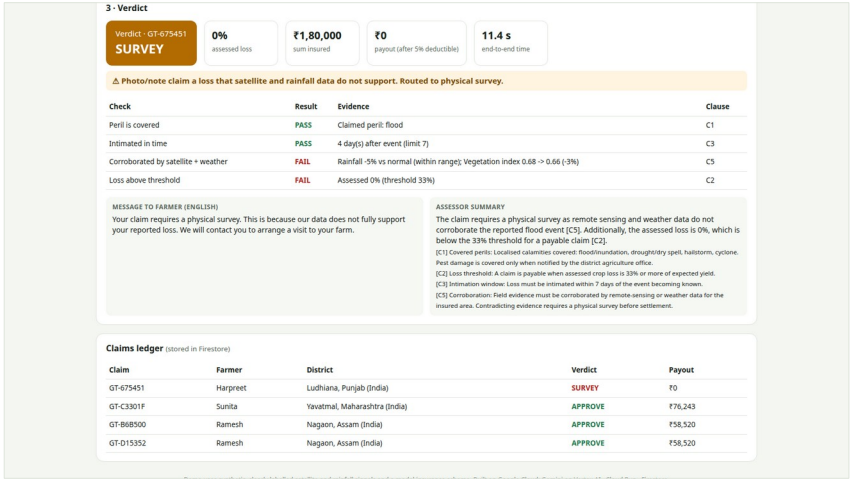
MESSAGE TO FARMER (HINDI)
आपकी फसल बीमा का दावा स्वीकृत हो गया है। आपको 58520 रुपये का भुगतान किया जाएगा। यह राशि जल्द ही आपके बैंक खाते में जमा कर दी जाएगी।

ASSESSOR SUMMARY
The claim is approved as the peril (flood) is covered [C1] and the assessed loss of 77% is above the 33% threshold [C2]. The intimation was timely [C3] and the field evidence was corroborated by remote sensing data [C5].
[C1] Covered perils: Localised calamities covered: flood/inundation, drought/dry spell, hailstorm, cyclone. Pest damage is covered only when notified by the district agriculture office.
[C2] Loss threshold: A claim is payable when assessed crop loss is 33% or more of expected yield.
[C3] Intimation window: Loss must be intimated within 7 days of the event becoming known.
[C4] Payout formula: Payout = Sum insured × assessed loss %, less a 5% deductible on the payout amount.
[C5] Corroboration: Field evidence must be corroborated by remote-sensing or weather data for the insured area. Contradicting evidence requires a physical survey before settlement.

Claims ledger (stored in Firestore)

Claim	Farmer	District	Verdict	Payout
GT-B6B500	Ramesh	Nagaon, Assam (India)	APPROVE	₹58,520
GT-D15352	Ramesh	Nagaon, Assam (India)	APPROVE	₹58,520

Demo uses synthetic, clearly labelled satellite and rainfall signals and a model insurance scheme. Built on Google Cloud: Gemini on Vertex AI · Cloud Run · Firestore.



3 · Verdict

Verdict · GT-675451
SURVEY

0% assessed loss
₹1,80,000 sum insured
₹0 payout (after 5% deductible)
11.4 s end-to-end time

⚠ Photo/note claim a loss that satellite and rainfall data do not support. Routed to physical survey.

Check	Result	Evidence	Clause
Peril is covered	PASS	Claimed peril: flood	C1
Intimated in time	PASS	4 day(s) after event (limit 7)	C3
Corroborated by satellite + weather	FAIL	Rainfall -5% vs normal (within range); Vegetation index 0.68 -> 0.66 (-3%)	C5
Loss above threshold	FAIL	Assessed 0% (threshold 33%)	C2

MESSAGE TO FARMER (ENGLISH)
Your claim requires a physical survey. This is because our data does not fully support your reported loss. We will contact you to arrange a visit to your farm.

ASSESSOR SUMMARY
The claim requires a physical survey as remote sensing and weather data do not corroborate the reported flood event [C5]. Additionally, the assessed loss is 0%, which is below the 33% threshold for a payable claim [C2].
[C1] Covered perils: Localised calamities covered: flood/inundation, drought/dry spell, hailstorm, cyclone. Pest damage is covered only when notified by the district agriculture office.
[C2] Loss threshold: A claim is payable when assessed crop loss is 33% or more of expected yield.
[C3] Intimation window: Loss must be intimated within 7 days of the event becoming known.
[C5] Corroboration: Field evidence must be corroborated by remote-sensing or weather data for the insured area. Contradicting evidence requires a physical survey before settlement.

Claims ledger (stored in Firestore)

Claim	Farmer	District	Verdict	Payout
GT-675451	Harpreet	Ludhiana, Punjab (India)	SURVEY	₹0
GT-C3301F	Sunita	Yavatmal, Maharashtra (India)	APPROVE	₹76,243
GT-B6B500	Ramesh	Nagaon, Assam (India)	APPROVE	₹58,520
GT-D15352	Ramesh	Nagaon, Assam (India)	APPROVE	₹58,520

Prototype Performance report/Benchmarking

Scenario	Expected	Verdict	Assessed loss	Payout (₹)	Latency (s)	Language
Flood, Assam (Hindi)	APPROVE	APPROVE	77.0%	58,520	13.8	Hindi
Drought, Maharashtra (Marathi)	APPROVE	APPROVE	52.8%	76,243	16.5	Marathi
Typhoon, Philippines (Filipino)	APPROVE	APPROVE	67.9%	61,925	14.4	Tagalog
Unsupported claim, Punjab	SURVEY	SURVEY	0.0%	0	15.0	English
Late report, Assam (20 days)	REJECT	REJECT	77.0%	0	19.9	English

Result: 5/5 verdicts correct on live Cloud Run · median end-to-end 15.0s per claim (vs weeks to months today) · 6/6 engine unit tests pass · identical inputs give identical decisions.

Additional Details/Future Development (if any)

Live data: Earth Engine Sentinel-2 NDVI per insured plot boundary, and gridded rainfall feeds.

Voice intake: Speech-to-Text so farmers can simply speak the claim.

Scheme knowledge: Vertex AI Search over each country's scheme documents, so policy clauses are retrieved and cited per programme.

Insurer console: survey queue for flagged claims, portfolio analytics in BigQuery.

Scale: tenant-per-insurer isolation, least-privilege IAM, regional deployment across JAPAC.

Note: satellite and rainfall values in this prototype are synthetic, clearly labelled sample signals; the scheme is a model area-yield scheme.

Provide links to your:

- GitHub Public Repository
- Demo Video Link (3 Minutes)
- Final Product Link

GitHub repository: https://github.com/lokithelearner/groundtruth

Demo video: https://groundtruth-673378516921.asia-south1.run.app/static/GroundTruth_Demo.mp4

Live prototype: https://groundtruth-673378516921.asia-south1.run.app

Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.

Thank you

